MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Cryptology – Lab_6

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1. Write a python code to perform One Time Pad

main.py

```python
import random

mode = input("Encrypt or decrypt (e/d): ").lower()
text = input("Enter text: ").upper().replace(" ", "")

if mode not in ("e", "d"):
    print("Mode must be e or d!")
    exit()
if not text.isalpha():
    print("Text must contain letters only!")
    exit()

if mode == "e":
    choice = input("Generate random key? (y/n): ").lower()
    if choice == "y":
        key = ""
        for i in range(len(text)):
            key += chr(random.randint(0, 25) + 65)
        print("Key:", key)
    else:
        key = input("Enter key: ").upper().replace(" ", "")
else:
    key = input("Enter key: ").upper().replace(" ", "")

if not key.isalpha():
    print("Key must contain letters only!")
    exit()
if len(key) != len(text):
    print("Key length must equal text length!")
    exit()

result = ""
for i in range(len(text)):
    p = ord(text[i]) - 65      # letter -> number (A=0)
    k = ord(key[i]) - 65
    if mode == "e":
        c = (p + k) % 26       # C = (P + K) mod 26
    else:
        c = (p - k) % 26       # P = (C - K) mod 26
    result += chr(c + 65)

print("Result:", result)
```

Output:

Encrypt or decrypt (e/d): e

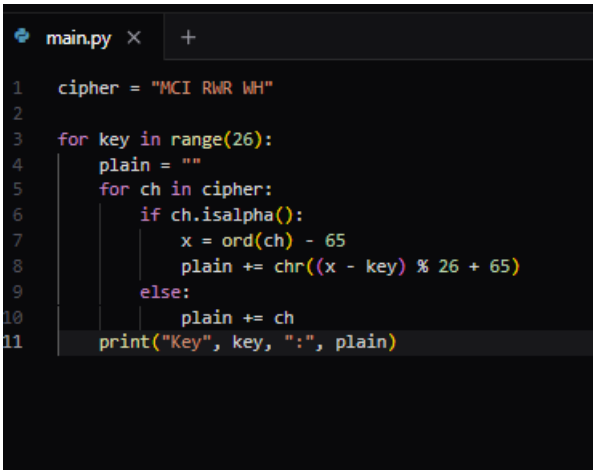
Enter text: Attackonce

Generate random key? (y/n): y

Key: CJIXTNYEAX

Result: CCBXVXMRCB

2. Write a Python code to break the shift cipher "MCI RWR WH" (Perform a brute-force attack by displaying all 26 possible plaintexts.)



Output:

Key 0 : MCI RWR WH
Key 1 : LBH QVQ VG
Key 2 : KAG PUP UF
Key 3 : JZF OTO TE
Key 4 : IYE NSN SD
Key 5 : HXD MRM RC
Key 6 : GWC LQL QB
Key 7 : FVB KPK PA
Key 8 : EUA JOJ OZ
Key 9 : DTZ INI NY
Key 10 : CSY HMH MX
Key 11 : BRX GLG LW
Key 12 : AQW FKF KV
Key 13 : ZPV EJE JU
Key 14 : YOU DID IT
Key 15 : XNT CHC HS
Key 16 : WMS BGB GR
Key 17 : VLR AFA FQ
Key 18 : UKQ ZEZ EP
Key 19 : TJP YDY DO
Key 20 : SIO XCX CN
Key 21 : RHN WBW BM
Key 22 : QGM VAV AL
Key 23 : PFL UZU ZK
Key 24 : OEK TYT YJ
Key 25 : NDJ SXS XI

3. Write a Python code to encrypt Rail Fence

main.py

```python
text = input("Enter text: ").replace(" ", "").upper()
try:
    rails = int(input("Enter number of rails: "))
except ValueError:
    print("Rails must be a whole number!")
    exit()

if text == "":
    print("Text cannot be empty!")
    exit()
if rails < 2 or rails > len(text):
    print("Rails must be between 2 and", len(text))
    exit()

fence = [""] * rails
row = 0
step = 1

for ch in text:
    fence[row] += ch
    if row == 0:
        step = 1           # at the top rail, go down
    elif row == rails - 1:
        step = -1          # at the bottom rail, go up
    row += step

for i in range(rails):
    print("Rail", i + 1, ":", fence[i])

print("Ciphertext:", "".join(fence))
```

Output:

Enter text: WEAREDISCOVERED

Enter number of rails: 3

Rail 1 : WECR

Rail 2 : ERDSOEE

Rail 3 : AIVD

Ciphertext: WECRERDSOEEAIVD